Fiscal Year:	FY 2018	**Task Last Updated:**	FY 07/13/2018
PI Name:	Jaing, Crystal Ph.D.		
Project Title:	International Space Station – Microbial Observatory of Pathogenic Virus, Bacteria, and Fungi (ISS-MOP) Project		
Division Name:	Human Research, Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	jaing2@llnl.gov	**Fax:**	FY
PI Organization Type:	GOVERNMENT	**Phone:**	925-424-6574
Organization Name:	Lawrence Livermore National Laboratory		
PI Address 1:	Physical & Life Sciences Directorate		
PI Address 2:	7000 East Ave,Mail Code L-452		
PI Web Page:			
City:	Livermore	**State:**	CA
Zip Code:	94550-9698	**Congressional District:**	15
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2014 Space Biology Flight (ILSRA) NNH14ZTT002N
Start Date:	05/01/2015	**End Date:**	12/31/2020
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sato, Kevin	**Contact Phone:**	650-604-1104
Contact Email:	kevin.y.sato@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 12/31/2020 per PI (Ed., 11/26/19) NOTE: End date changed to 12/31/2019 per F. Hernandez/ARC (Ed., 4/18/19) NOTE change in grant number and end date per NSSC information and F. Hernandez/ARC; note grant 80NSSC18K0113 has period of performance of 10/20/2017-4/30/2019 (Ed., 1/18/2018) NOTE: Extended to 3/31/2019 per F. Hernandez/ARC (Ed., 8/22/17)		
Key Personnel Changes/Previous PI:	No changes.		
COI Name (Institution):	Mehta, Satish Ph.D. (Wyle Laboratories, Inc.) Pierson, Duane Ph.D. (NASA Johnson Space Center) Smith, David Ph.D. (NASA Ames Research Center) Venkateswaran, Kasthuri Ph.D. (NASA Jet Propulsion Laboratory)		
Grant/Contract No.:	80NSSC18K0113 ; NNX15AJ29G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	[Ed. note (January 2018): change in grant number and end date per NSSC information and NASA Ames Research Center; note grant 80NSSC18K0113 has period of performance of 10/20/2017-4/30/2019] International Space Station – Microbial Observatory of Pathogenic Virus, Bacteria, and Fungi (ISS-MOP) project will generate the microbial censuses associated with various ISS modules using traditional culture-based methods and state-of-the art molecular techniques. The objectives of the ISS-MOP project are to measure presence of viral and select bacterial and fungal pathogens and correlate their presence on crew. The effects of the spaceflight environment on viral and microbial pathogen dynamics are largely unknown and represent both a significant gap in our knowledge and an important opportunity to study the presence of microbial populations and predict health risks during long-term space exploration. The hypothesis of the ISS-MOP project is that certain short-living viruses associated with inert environmental spaces in the closed habitat of the ISS can have an adverse influence on crew health. To test this hypothesis the team will catalog viruses associated with surfaces and air of the ISS modules using the all-encompassing Lawrence Livermore Microbial Detection Array (LLMDA), which contains probes to detect not only viruses but also clinically important human bacterial and fungal pathogens. In addition, skin and saliva samples from crewmembers will be requested to test the hypothesis. There are three aims proposed in this ISS-MOP project. While the cultivation-based assay (Aim 1) provides a valid estimation of microbial presence, it offers only a limited assessment of the phylogenetic/pathogenic diversity and physiological breadth of the microbial population present. Therefore, the team will use the novel and comprehensive LLMDA microarray, which is specifically designed to elucidate the pathogenic viral and microbial diversity profiles of complex samples to at least the species level (Aim 2). In addition, we will characterize the virulence and antibiotic resistance mechanisms associated with these pathogens from the ISS by developing a new Virulence and Antibiotic Resistance gene array and analyze the samples on this array. The final deliverable of the proposed project to NASA will be the ISS-MOP dataset, a compilation of all of the genomic sequences and genetic information of viruses and microbes encountered on and within the ISS habitat (Aim 3). The objectives of the ISS-MOP project directly respond to the National Research Council Committee for the Decadal Survey on Biological and Physical Sciences in Space recommendation P1 where it states, “to establish a microbial observatory program on the ISS to conduct long-term, multigenerational studies of microbial population dynamics.” The ISS-MOP project will also provide answers to NASA Space Biology Guiding Questions: How do alterations in gravity affect in microbial cells (CMM1); Does the enhanced virulence or other genetic changes observed with microbial cultures flown in space occur in other species (CMM4); and What are the genetic changes underlying physiological responses to spaceflight demonstrated in biological systems (CMM5).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The effects of the spaceflight environment on viral and microbial pathogen dynamics are largely unknown and represent both a significant gap in our knowledge and an important opportunity to study the presence of microbial populations and predict health risks during long-term space exploration. International Space Station – Microbial Observatory of Pathogenic Virus, Bacteria, and Fungi (ISS-MOP) project will generate the microbial censuses associated with crewmembers and various ISS modules using both traditional culture-based methods and state-of-the art molecular techniques to a degree that has not previously been possible.
Task Progress:	Summary of accomplishments: Flight and sample status: 1). Successfully enrolled four crewmembers to carry out our studies to collect ISS environmental surface wipes, air samples, crew body swabs, and saliva samples for microbiological and viral analysis; 2). ISS environmental samples have been collected over the performing period. Crewmember body swab and saliva samples from preflight, infight, and postflight samples are complete for the first crewmember. Sample collection for crewmembers 2-4 is still ongoing. Progress towards scientific goals 1. Cultivation-based assay to estimate microbial presence. We have performed microbiological culture of environmental surface wipe and air samples from the first three flights (F4, F5, F6). We started our first flight as F4 as a continuation of the flight name from Microbial Tracking-1 (MT-1, PI: Kasthuri Venkateswaran) project which collected samples from three flights (F1, F2, F3). The surface wipes were plated in three different agar plates, Blood Agar (BA), Reasoner's 2A Agar (R2A), and Potato Dextrose Agar (PDA). The bacterial (grown from BA and R2A) and fungal (grown from PDA) contents from the samples were evaluated using Colony Forming Units (CFU) /m2. Overall, we observed bacterial and fungal growth from surface samples from all three flights. The microbial growth concentration ranged from 10^2 to 10^6 CFU/m2. The culture of air samples did not yield significant growth as compared to negative control samples possibly due to the much lower biomass available from air filters. 2. Molecular analysis to characterize ISS environmental and crew microbial profiles. We performed the Lawrence Livermore Microbial Detection Array (LLMDA) analysis on F4 environmental surface wipes and all of crewmember #1 saliva and body swab samples collected pre-, in-, and post-flight. In total, ~100 samples and controls have been subjected to LLMDA analysis. We also performed metagenomic DNA sequencing of the first two environmental samples (F4 and F5) and the crew 1 samples. We performed bioinformatic analysis of F4, F5, and crew 1 samples to determine any commonality and differences between two flights, between crew and environmental microbiome, and between preflight, inflight, and postflight to determine any diversity changes during spaceflights. We observed that Propionibacterium, Staphylococcus, Rothia, Veillonella, Prevotella are among the top bacteria detected in the samples. The microbial diversity between the three datasets also showed some differences. The last sample collected for crewmember #1, R+180 days, was not included in these results due to time constraints. We are preparing manuscripts to report these findings. 3. ISS-MOP dataset, a compilation of all the genomic sequences and genetic information of viruses and microbes encountered on and within the ISS habitat. We will compile all data generated to date to make this into a dataset and provide to the NASA Data Archive.

	Team interactions: 1). Team conducted monthly conference calls to discuss progress and results and operational matters during this entire performing period; 2). ISS-MOP investigators conducted an in-person meeting on June 18, 2018 at Lawrence Livermore National Laboratory (LLNL) to discuss work progress, lessons learned, and publication plans; 3). Team members attended the American Society for Gravitational and Space Research (ASGSR) meeting in Seattle in Oct 2017 to review work progress and present preliminary data. The PI attended the Human Research Program (HRP) meeting in Houston, TX in January 2018 and presented a poster on the ISS-MOP project.
Bibliography Type:	Description: (Last Updated: 04/07/2023)
Articles in Peer-reviewed Journals	Urbaniak C, Sielaff AC, Frey KG, Allen JE, Singh N, Jaing C, Wheeler K, Venkateswaran K. "Detection of antimicrobial resistance genes associated with the International Space Station environmental surfaces." Sci Rep. 2018 Jan 16;8(1):814. https://doi.org/10.1038/s41598-017-18506-4 ; PubMed PMID: 29339831; PubMed Central PMCID: PMC5770469 , Jan-2018
Articles in Peer-reviewed Journals	Venkateswaran K, Checinska Sielaff A, Ratnayake S, Pope RK, Blank TE, Stepanov VG, Fox GE, van Tongeren SP, Torres C, Allen J, Jaing C, Pierson D, Perry J, Koren S, Phillippy A, Klubnik J, Treangen TJ, Rosovitz MJ, Bergman NH. "Draft genome sequences from a novel clade of Bacillus cereus Sensu Lato strains, isolated from the International Space Station." Genome Announc. 2017 Aug 10;5(32):e00680-17. https://doi.org/10.1128/genomeA.00680-17 ; PubMed PMID: 28798168; PubMed Central PMCID: PMC5552977 , Aug-2017
Articles in Peer-reviewed Journals	Be NA, Avila-Herrera A, Allen JE, Singh N, Checinska Sielaff A, Jaing C, Venkateswaran K. "Whole metagenome profiles of particulates collected from the International Space Station." Microbiome. 2017 Jul 17;5(1):81. (Erratum in: Microbiome. 2017 Sep 1;5(1):111.) https://doi.org/10.1186/s40168-017-0292-4 ; PubMed PMID: 28716113; PubMed Central PMCID: PMC5514531 , Jul-2017
Awards	Jaing C, Slezak T, McLoughlin K, Thissen T, Be N, Gardner S. "R&D100 award for LLMDA, November 2017." Nov-2017